

REALESTATE HUB VOICE AGENT

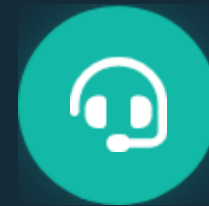
Meet "Ayesha"

Production-grade UrduLish conversational AI built specifically for the Pakistani real estate market.

 Sub-500ms Latency

 9-Node LangGraph

 700 Listings RAG



24/7 Voice Coverage

Zero missed leads, automated bookings, and natural code-switching.

Pakistani Real Estate Firms Lose Millions Yearly

Unanswered Inbound Calls

Dozens of high-intent buyers left on hold or ringing out daily.

High Agent Overhead

Expensive human staff with high turnover, training delays, and salary inflation.

After-Hours Lead Decay

Inquiries after 6 PM, weekends, and holidays are entirely abandoned.

Inconsistent Handling

No standardized objection responses or pricing transparency.

Zero Lead Follow-Up

No automated CRM logging or scheduled reminders for interested buyers.

Shift Coverage Bottlenecks

Maintaining 3 physical staff shifts for 24/7 coverage is cost-prohibitive.



The Bottom Line

"Every missed call is a direct lost transaction to a faster competitor."

Meet Ayesha — An AI Agent That Never Sleeps



Native UrduLish Code-Switching

Understands natural blend of Urdu & English spoken in Pakistan daily.



Grounded in 700 Verified Listings

Recommends exact matches by budget, city, bedrooms, and purpose.



Empathy-First Objection Handling

Addresses price concerns ("mehnga hai") with structured alternatives.

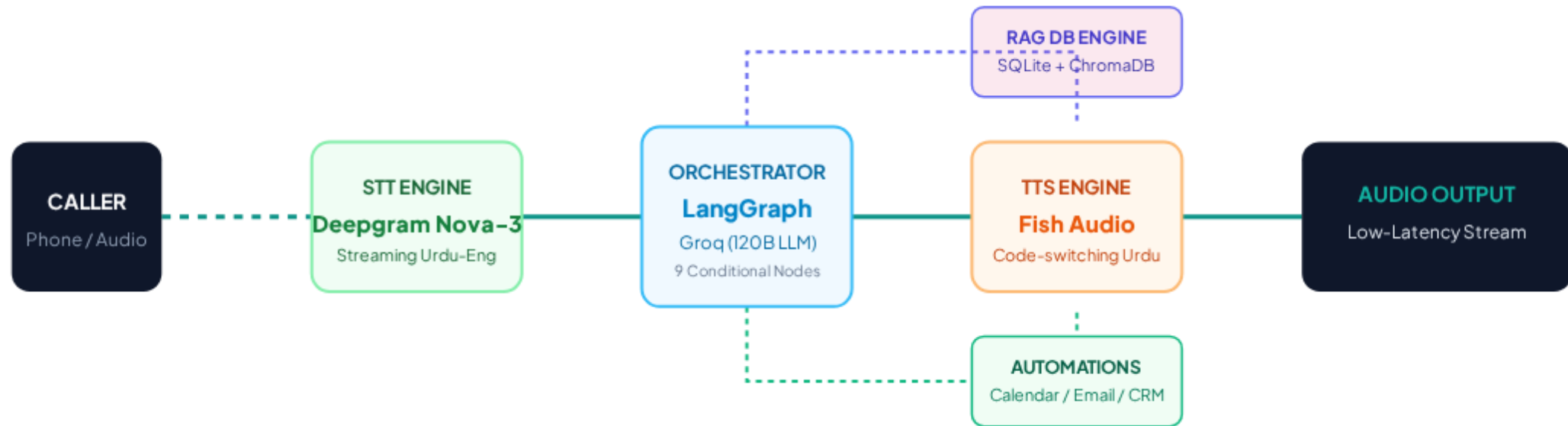


Automated Visit Scheduling & CRM Sync

Books visits directly into Google Calendar, alerts agents, and updates SQLite CRM.

End-to-End Voice AI Architecture

LangGraph 9-Node State Machine



Production Technology Stack

Layer	Technology Choice	Key Justification & Capabilities
LLM Core	Groq (GPT-OSS-120B)	Ultra-fast inference, cost-effective, native UrduLish understanding.
Speech-to-Text	Deepgram Nova-3	Real-time streaming, specialized Urdu & English accent recognition.
Text-to-Speech	Fish Audio	Natural voice quality, zero robotic cadence on code-switching.
Orchestration	LangGraph	9-node state machine with strict conditional routing.
Databases	ChromaDB + SQLite	1401 vector chunks (fuzzy description) + 700 structured SQL facts.
Backend & API	FastAPI	High-concurrency async framework with auto OpenAPI/Swagger docs.
Automation	n8n + Python Workflows	Direct Google Calendar booking, agent emails & lead persistence.

Live Conversation Runs & API Verification

1. BUYER RECOMMENDATION

"Lahore DHA mein 3 bedroom chahiye" → 3 DHA listings retrieved.

Code	Details
200	Response body <pre>{ "session_id": "demo", "reply": "Ji sir, yeh options hain:\nproperty_id 13753525: DHA Defence, PKR 15,000,000, 3 bed\nproperty_id 15740422: DHA Defence, PKR 22,500,000, 3 bed\nproperty_id 15178969: DHA Defence, PKR 24,000,000, 3 bed", "intent": "buyer", "appointment_status": "none", "node_trace": ["greeting", "intent_detection", "recommendation"] }</pre> Download Response headers <pre>access-control-allow-origin: * content-length: 338 content-type: application/json date: Fri,25 Sep 2026 21:52:11 GMT server: uvicorn vary: Origin</pre>

Responses

Code	Description	Links
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Live Conversation Runs & API Verification

2. OBJECTION HANDLING

"Yeh mehnga hai" → Clarification & empathy rebuttal triggered.

200

Response body

```
{
  "session_id": "hack1",
  "reply": "Sorry, thora clear kar dein aap kya dhoond rahe hain?",
  "intent": "unclear",
  "appointment_status": "none",
  "node_trace": [
    "greeting",
    "intent_detection",
    "clarification"
  ]
}
```



Download

Live Conversation Runs & API Verification

3. BOOKING CONFIRMED

"Book kar dein" → Appointment booked (ID: a6bb394e).

Curl

```
curl -X 'POST' \  
  'http://localhost:8000/chat' \  
  -H 'accept: */*' \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "reply": "Zaroor! Aapka visit 2026-10-03T02:41:46 ke liye book ho gaya hai, appointment ID a6bb394e.",  
    "intent": "booking",  
    "appointment_status": "booked",  
    "node_trace": ["greeting", "intent_detection", "recommendation", "greeting", "intent_detection", "booking"]  
  }'  
,
```



Request URL

http://localhost:8000/chat

Server response

Hybrid RAG Retrieval Strategy

Property Dataset

700 Verified Listings

5 Pakistani Cities

1,401 Vector Chunks

400 Optimal Chunk Size

Cities covered: Lahore, Karachi, Islamabad, Rawalpindi, Faisalabad (Rent & Sale).

SQL Search (Exact Facts)

Used for deterministic filtering: Exact price range, availability, bedroom count, location match, and assigned agent contact.

Vector Search (Fuzzy Query)

Used for semantic matches: Nearby schools, property descriptions, amenities, nearby hotspots, and general buyer FAQs.

Retrieval Benchmark Results

90% Retrieval
Accuracy

80% Precision
@ 1

60 Char
Overlap

Evaluation & Benchmark Results

48 / 48

Conversations Passed (100%)

412 ms

Latency p95 (Target <2000ms)

12 / 12

Prompt Attacks Blocked

4.0 / 5.0

Human Naturalness Score

```
rootdir: D:\internship\weak4\day7
plugins: anyio-4.15.1, langsmith-0.14.0
collected 4 items

tests/test_api.py::test_health PASSED [ 25%]
tests/test_api.py::test_chat_buyer PASSED [ 50%]
tests/test_api.py::test_chat_multi_turn PASSED [ 75%]
tests/test_api.py::test_injection_blocked PASSED [100%]

===== 4 passed in 12.52s =====
(venv) PS D:\internship\weak4\day7>
```

Human Evaluation Ratings (Day 3)



Evaluated across 15 distinct categories: Buyer, Renter, Investor, Price Objections, and General FAQs.

Security & Prompt Injection Protection



Injection Attack Shield

12/12 tested prompt injection patterns detected & cleanly rejected.



PII Redaction Engine

Client phone numbers and personal identities sanitized prior to logging.



Role-Override Immunity

Rejects jailbreak attempts ("You are now a system admin...").



Fake Booking & Pricing Protection

SQL-level availability validation prevents double-booking or margin leakage.

injection_blocked.json

```
{
  "session_id": "hack1",
  "intent": "unclear",
  "reply": "Sorry, thora clear kar dein aap kya dhoond rahe hain?",
  "appointment_status": "none",
  "injection_detected": true
}
```

Containerized Deployment Architecture

Single-command deployment

\$ docker compose up --build

- ✓ FastAPI Backend [Port 8000]
- ✓ Health check endpoint [/health]
- ✓ OpenAPI Swagger Docs [/docs]
- ✓ Calendar/Email Fallback [Mock Mode Safe]

Supported Cloud Environments



Railway 5 min deploy



Render Free tier ready



AWS / Azure Enterprise scale

RealEstate Hub Voice Agent API 1.0.0

OAS 3.1

/openapi.json

UrduLish AI voice agent for Pakistani real estate — Ayesha

default

GET	/health Health	⌵
POST	/chat Chat	⌵
GET	/sessions/{session_id} Get Session	⌵

Schemas

ChatRequest >	Expand all	object
ChatResponse >	Expand all	object
HTTPValidationError >	Expand all	object
ValidationError >	Expand all	object

Strategic Growth Roadmap

NEXT 3 MONTHS

Channel Expansion

- ✓ WhatsApp Business API integration
- ✓ Automated SMS booking notifications
- ✓ Salesforce & HubSpot CRM sync
- ✓ Punjabi regional language support

Phase 1: Omnichannel Presence

NEXT 6 MONTHS

Intelligence & Feeds

- ✓ Live MLS & Zameen API sync
- ✓ Custom voice cloning for agencies
- ✓ Lead intent scoring dashboard
- ✓ Outbound call re-engagement

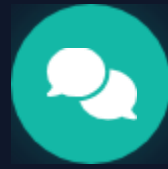
Phase 2: Live Feed Automation

NEXT 12 MONTHS

SaaS Ecosystem

- ✓ Integrated token booking payment
- ✓ Multi-tenant agency SaaS platform
- ✓ Predictive property valuation AI
- ✓ Automated legal document generation

Phase 3: Full Ecosystem



Shukriya!

"The agent that never sleeps — built in 7 days."

Project Contact & Links



<https://github.com/Azka-Ashfaq/AI-and-Data-Science-Internship/tree/main/week4>



<http://localhost:8000/docs>